

Mingtai Zhang

zhangmt@st.gxu.edu.cn | taitaile071441@gmail.com

Education

Guangxi University | School of Resources, Environment and Materials

2022 – 2026 (expected Jun.)

B.Eng. in Materials Science and Engineering

Coursework: physical chemistry, fundamentals of materials science, physical metallurgy, principles of metallurgical processes, materials characterization methods, electrical and electronic engineering, mechanical design.

My quantitative and computational training is self-taught. The computational work below was done with a research group at Northeastern University (Qinhuangdao), in parallel with a full experimental thesis in electrochemistry.

Research Summary

Across five projects in four fields I have held the same role: I build the part that decides whether a claim is true. In my first-author work that meant defining an attribution boundary in conversational memory, and showing that training which improves a model's replies can corrupt what it writes to a user's persistent profile, a failure reply-only evaluation never sees. Elsewhere it has meant the evaluation harness for an identifiability result in coherent imaging, and the full experimental verification of a minimax bound on estimating a selected system's accuracy from a fixed label budget. The habit began in electrochemistry, where every claim about the interface mechanism had to survive multiple independent measurement channels. I want to work on measurement design for systems that cannot be observed directly: what an evaluation has to cover, and what it costs, before a reported improvement can be believed.

Research Interests

Evaluation and auditing of LLM systems | memory, attribution, and trustworthy human–AI interaction | LLM agents for evidence-grounded reasoning | inverse problems and sample complexity | AI for experimental science

Publications and Manuscripts

Co-author lists for papers under double-blind review are withheld until notification, following the anonymity policy of each venue.

Under Review

- [1] Paper on attribution boundaries in conversational AI memory (title withheld during review) **First author**; under double-blind review, a top-tier HCI conference.
- [2] Paper on the accuracy–evaluation trade-off in test-time selection (title withheld during review) **Second author**; under double-blind review, a top-tier machine-learning conference.
- [3] Yuchen Miao, **Mingtai Zhang**, Zijun Wang, Ruobing Han, Chang Han, and Siyang Xu. “Two-Scan Recovery of Sparse Defects in Ptychography.” **Second author**; under review, *IEEE Signal Processing Letters*, 2026. (CRediT: data curation, software.)

Accepted

- [4] Yuchen Miao, Zijun Wang, Chang Han, Yurui Shi, **Mingtai Zhang**, and Siyang Xu. “The ‘10th Juror’: Open-Set Standpoint Screening for Bureaucratic Bias Detection.” *EMNLP 2026* (Main Conference) [CORE A*; CCF-B].
- [5] Guang Zhang, Dong Yang, Kewang Yang, Farwa Mushtaq, Meng Zhang, **Mingtai Zhang**, Jieshang Lu, Yucheng Wen, and Meinan Liu. “Modified Ether Electrolyte by Siloxane Additive with Abundant Si–O Bonds for High-Voltage Lithium Metal Batteries.” *Journal of Physics: Conference Series* **3165**, 012009 (2025); CEAMS 2025 [EI-indexed]. DOI: 10.1088/1742-6596/3165/1/012009.

Research Experience

Attribution and Memory in Conversational AI Agents

2026

First author, under double-blind review | problem formulation, measures, experiments, writing

- **Problem.** When a user role-plays a fictional character, a companion agent can store the character's statements as facts about the real person. I formalized *boundary non-interference*: a proposition supported only inside a fictional frame must not be asserted as a fact about the real user without attribution.
- **Two output surfaces, two measures.** Instead of scoring overall response quality, I separated the conversational channel from the persistent-memory channel, with a cross-frame leakage rate for replies and a profile-write corruption rate for generated profiles. Each came with explicit scoring rules and coverage diagnostics, plus a secondary relational-positioning construct kept clearly subordinate.
- **Dual ground truth.** Built 512 rule-generated Chinese episodes that carry paired fictional and real-user values for the same attribute, so every error is attributable rather than inferred. Scored 40,757 replies and an 80-condition factorial of 25,600 generated profiles across boundary cues, model scale, thinking settings, and writer prompts.
- **Main finding.** Two supervised fine-tuning variants improved conversational attribution without consistently improving profile writing, and conversation-targeted fine-tuning raised observed write corruption in all eight reported condition summaries. Reply-level evaluation alone would have scored both interventions as successes.

Accuracy and Evaluation Cost in Test-Time Selection

2026

Second author, under double-blind review | designed and ran the entire experimental verification

- Selecting a better response also makes its accuracy harder to measure, because the favored answers are rare in the labels you already have. The paper characterizes the best accuracy reachable with N candidates when evaluation reuses L labels from the base generator, and shows that every continuous-score best-of- N rule has worst-case audit risk $\Theta(1 \wedge N/L)$, so holding precision fixed costs labels in proportion to the candidate budget.
- The two constraints, which output laws a candidate pool can reach and how hard each law is to audit, define an accuracy–audit frontier whose dual discounts correctness in regions with poor label coverage. Cross-fitted frontier policies on four frozen reasoning banks (GSM8K, MATH, CodeContests, MiniF2F) beat uniform best-of- N mixtures by 2.92 points at matched audit budgets, and match fully optimized order-statistic and quadratic baselines.

Sample Complexity of Sparse Recovery in Coherent Imaging

2026

Second author, IEEE SPL (under review) | forward-model simulation and the full evaluation harness

- The paper asks a minimum-measurement question: given a known uniform background carrying N unknown defects, how many scan positions are needed for unique recovery? The answer is two, and one is provably insufficient because of a conjugate-reflection ambiguity. Working on an identifiability result changed how I approach evaluation in general, in that I now ask whether the measurement can determine the answer before asking which method wins.
- I handled data curation and software: the forward-model simulation and the whole evaluation harness, including Gaussian probes, scan geometries, sparse-defect ensembles, and detector-noise injection, then benchmarked the proposed $2N$ -lag decoder against full-data linearization and the iterative phase-retrieval baselines PIE and HT-WF over defect-separation and SNR sweeps.

Open-Set Standpoint Screening with LLM Agent Panels

2026

Fifth author, EMNLP 2026 Main Conference | evaluation-set construction and experimental analysis

- Contributed evaluation-set construction and experimental analysis to MARS-Gov, which drops the closed-world assumption behind fixed bias taxonomies. A Scout proposes affected groups that no taxonomy defines, fixed and dynamic Jurors deliberate over retrieved legal and policy evidence, a Judge verifies, and a restoration agent rewrites while preserving meaning. The system reached $F1 = 0.880$, 20.2 points above the strongest zero-shot LLM, and leave-one-category-out evaluation showed that genuinely unseen categories are recoverable rather than memorized.

Controlled Electrochemistry | undergraduate thesis, Guangxi University

2025 – 2026

Sole author, advisor Prof. Meinan Liu | experimental design, three measurement channels, all analysis

- Ran a single-variable controlled study on lithium-metal anode interfaces, with every quantity fixed except one process parameter and an unmodified baseline carried through as a parallel control. With no direct image of the interface, every claim had to come from agreement between independent measurements.
- Used three cell configurations as three independent measurement channels: half cell, symmetric cell, and full cell. A claim that survived in only one channel could not be reported as a general improvement. Nucleation overpotential fell from 145 to 114 mV, symmetric cells stayed near 30 mV for 330 h, and full cells kept 85.3% after 600 cycles.
- Localized failure on both sides of the optimum instead of reporting only the best setting: too little treatment leaves uncovered sites and collapses at cycle 45, too much builds a layer that cracks and collapses at cycle 30. Measured the mechanism separately by impedance spectroscopy and X-ray diffraction, so the performance claim rests on an identified cause and not a correlation. Kept the whole route on commercial reagents and standard hardware so that other labs can reproduce it.

Systems Built and Deployed

Kangjie Returning-Student Tutoring Platform, Yuncheng, Shanxi | Founder

Winter 2026

- Founded and ran a non-profit tutoring service staffed by university students home for the holiday, owning scheduling, tutor–student matching, progress tracking, and daily operations.
- Rebuilt the operating workflow around AI agent tooling and automated the coordination work that had been manual, which cut the staffing cost of running the service substantially. It was also a practical test of whether agents hold up as operators rather than demos; in total the service mobilized 18 tutors to serve 37 secondary-school students.

On-Demand Fitness Coaching Platform, Guangxi University | Co-Founder

2026 – present

- Co-founded a live two-sided service with the Guangxi University Rehabilitation Association, matching certified student coaches to users for one-on-one on-site sessions. I own operating strategy, supply coordination, and the commercial model.

Industry Grounding

Sixteen manufacturing enterprises across six industries, Guangxi | four supervised placements

2024 – 2026

- Traced the aluminium chain end to end, from bauxite mining and Bayer refining through electrolysis, casting, extrusion, rolling, and lithium-battery foil, plus integrated steelmaking, automotive stamping-to-assembly, construction machinery, photovoltaic module lines, and superhard-material sintering. I have stood at the inspection points where industrial process data is actually produced, and it shapes how I think about deployed models: on a plant floor a system is judged by throughput, yield, and inspectability, not by one headline number.
- **Closest to my own work:** at **Nanning Industrial Investment Aluminium-Based New Materials Group** I followed the full ingot-to-foil chain for 12 μm lithium-battery cathode current-collector foil. The component I hand-assemble into coin cells is made there by the hundred-thousand tonne.

Leadership, Honors, and Technical Skills

Leadership: Captain, Guangxi University Football Team | Youth League Branch Secretary, University Football Association | Chair, Student Representative Assembly, School of Resources, Environment and Materials.

Honors: Silver Award, Guangxi University Career Planning Competition (2023) | Third Prize, Guangxi University Metallography Contest (2023) | First Place (2025) and Third Place (2024), Guangxi “Qianli Cup” Intercollegiate Football League, undergraduate division.

Programming & Tools: Python, PyTorch, $\text{\LaTeX}$, MATLAB, C, Claude Code, OpenClaw, Origin | **ML:** supervised fine-tuning of open-weight LLMs, large-scale batched inference, LLM-as-judge pipelines, RAG, multi-agent systems, checkpoint selection and validity filtering | **Evaluation:** factorial design, parallel controls, replication arms, holdout construction, coverage diagnostics, failure localization, pre-stated validity limits | **Signal & Imaging:** forward-model simulation, phase retrieval (PIE, HT-WF), sparse recovery, sample-complexity benchmarking | **Instrumentation:** electrochemical workstations, EIS, XRD, SEM, XPS, glovebox